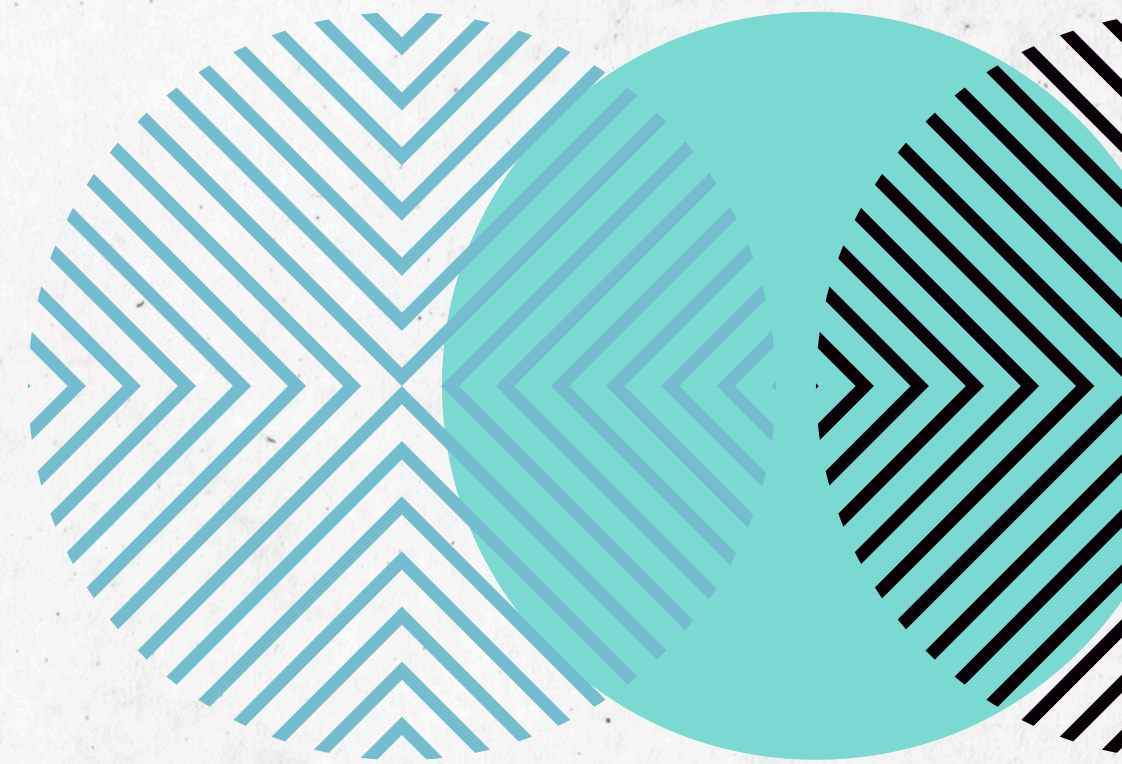


Binary Addition Project with Arduino



Yuri Chaves 10/09





Introduction

- Reading the Arduino introduction documentation
- Creating the GitHub repository
- Adding the base code provided by the professor
- Commenting the code in English
- Implementing full adder logic

The project started with reading the Arduino documentation, followed by creating a GitHub repository. The base code was added and commented in English to improve clarity. The full adder logic was implemented using two main functions: `somaBit` and `somaCarryBit`.



Software Demo

- Digital inputs: pins 0 to 7
- Control button: pin 13
- Visual outputs: LEDs on pins 8 to 12
- Addition of two 4-bit binary numbers

The system reads two 4-bit binary numbers through switches connected to pins 0 to 7. The button on pin 13 activates the addition operation. The result is displayed on LEDs connected to pins 8 to 12, representing the sum bits and the carry bit.

Survey of what was developed

- Functional code with full adder logic
- Explanatory comments in English
- Public GitHub repository
- PDF presentation

I developed a functional system that performs binary addition with carry. The code was commented in English and is available on GitHub. The presentation was structured to demonstrate all stages of the project.



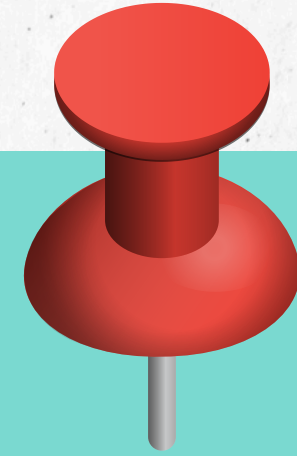
Future development activities

- Implement binary subtraction
- Add 7-segment display
- Create graphical interface using Processing

I plan to implement binary subtraction, add a 7-segment display for better visualization, and create a graphical interface using Processing to make the project more interactive.

Project License

The project is licensed under the MIT License, allowing use, modification, and distribution with attribution.



I chose the MIT license because it's simple and permissive, allowing other developers to freely use and modify the project as long as credit is given.

The image features a light gray, textured background. In the top-left and bottom-right corners, there are decorative circular patterns. These patterns consist of concentric circles with different line styles: the outermost ring has a light blue chevron pattern, the middle ring is a solid teal color, and the innermost ring has a black chevron pattern. The text "Thank you for your attention!" is centered in the middle of the page in a dark gray, sans-serif font.

Thank you for your
attention!